

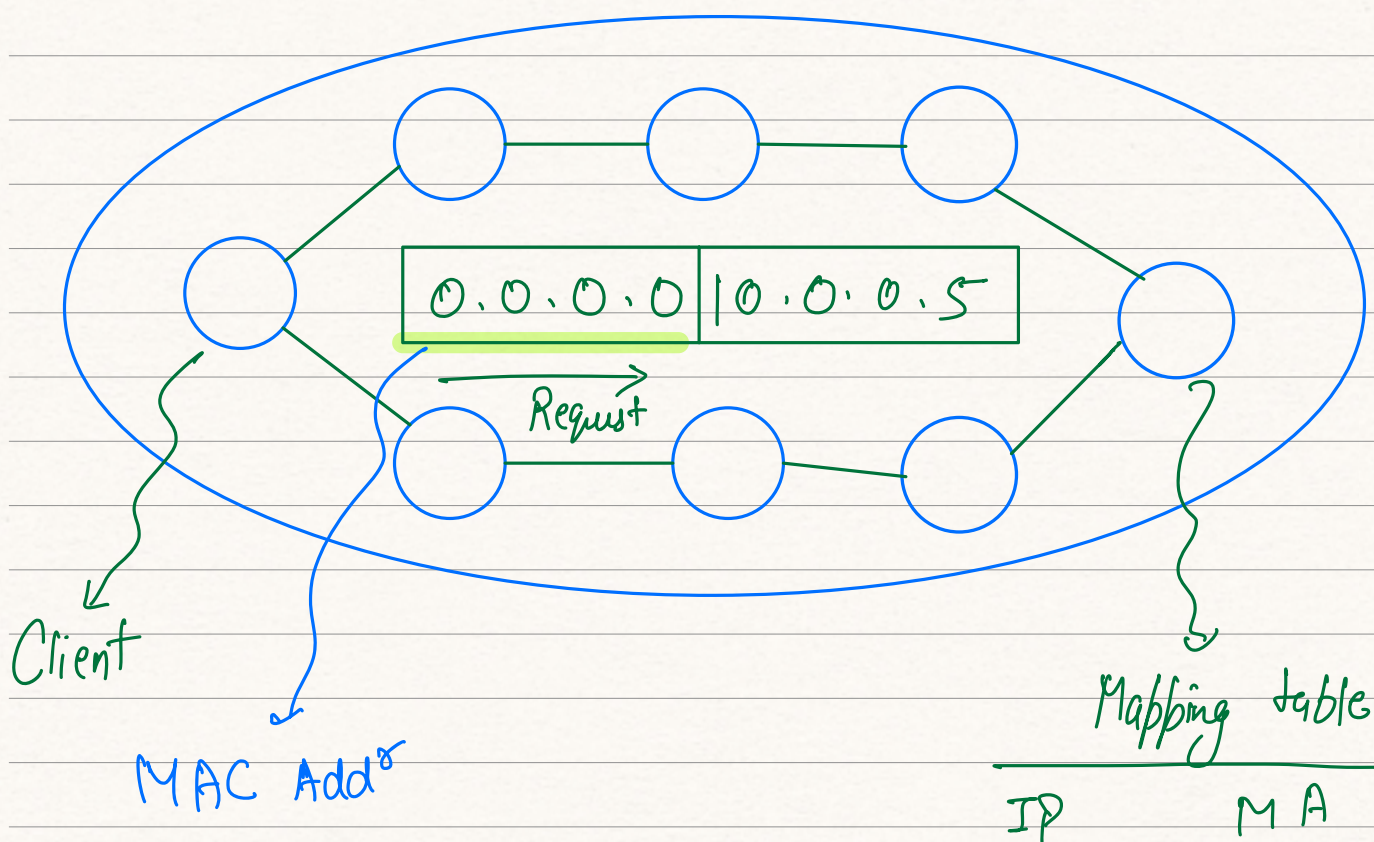
Subject: / /

Private IP

$\left\{ \begin{array}{l} 10.0.0.0 \text{ --- } 10.255.255.255 \\ 172.16.0.0 \text{ --- } 172.31.255.255 \\ 192.168.0.0 \text{ --- } 192.168.255.255 \end{array} \right\}$

* NAT (Network address Translation)

Case I :- Assigning IP Address in a LAN



14:1A:1B:1C:1D:1E

10.0.0.7

14:1A:1B:1C:1D:1E

Subject:

SIP	DIP
10.0.0.5	255.255.255.255

Stateful

1. Once the server is loaded with the new operating system, it will get the group of IP, out of which one IP is assigned to server,
2. The server IP is informed to all the client using LBA (Limited Broadcast Address)
3. Every computer will put the request ^{with} 0.0.0.0 as a source address, along with MAC address is also transmitted, so server can understand which computer has putted request
4. When a server has assigned an IP address, it maintains a mapping table b/w the IP address and MAC address

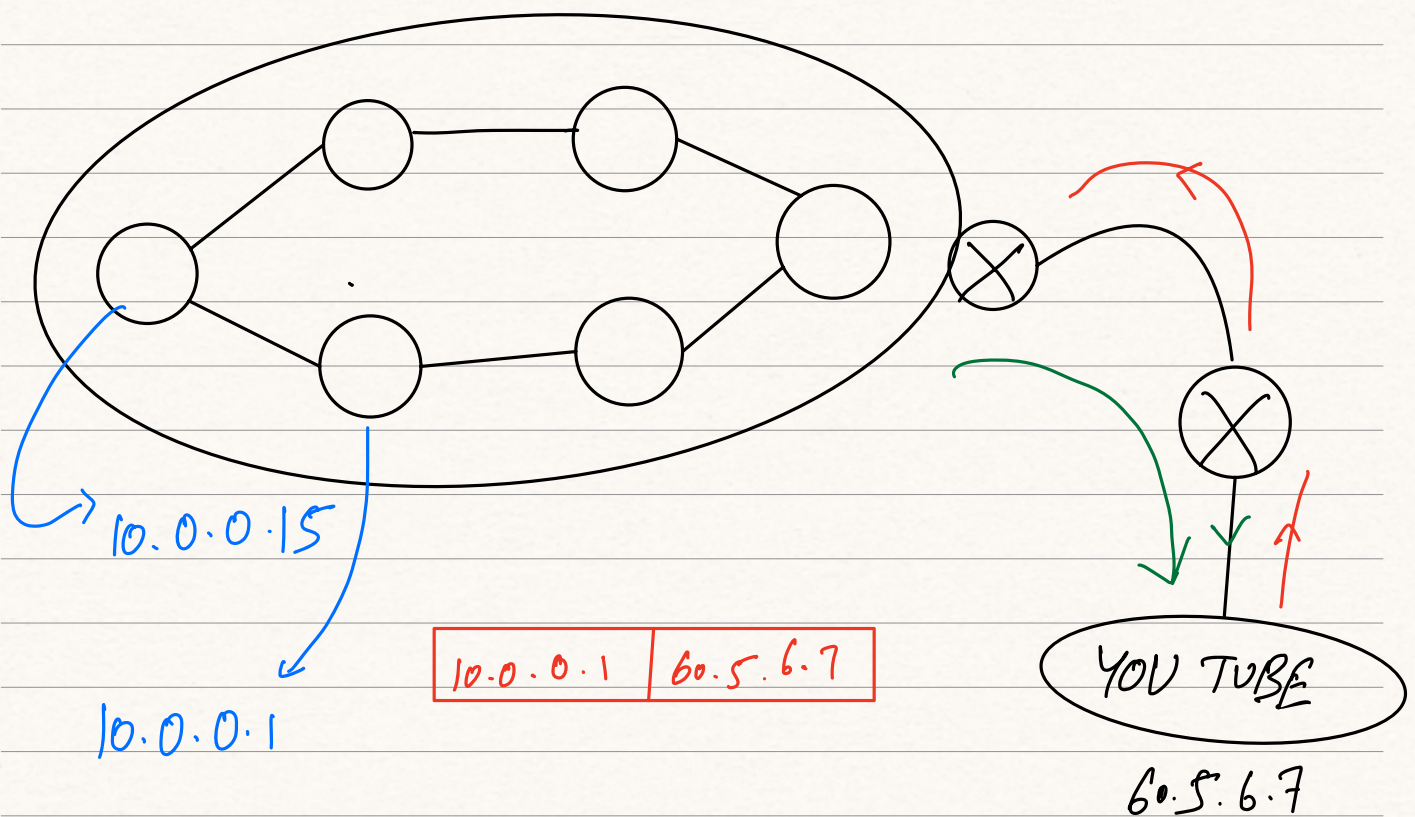
* Any computer which is not having any IP address still it can transmit the data using 0.0.0.0 as the source IP address.

Subject:

1. It is a **stateful** protocol becz it store a server side in the form of mapping table.

2. In IPv4 assigning these IP add^r, it supports stateful protocol

(II) Assigning Public IP (Internet Service)



<http://www.youtube.com>



Subject: / /

1. NAT router convert private IP into public IP, when the packet is going out of the network, it convert public IP into private IP, when the packet coming inside the network.
2. Then public IP addresses are effectively utilised with the help of private IP addresses.